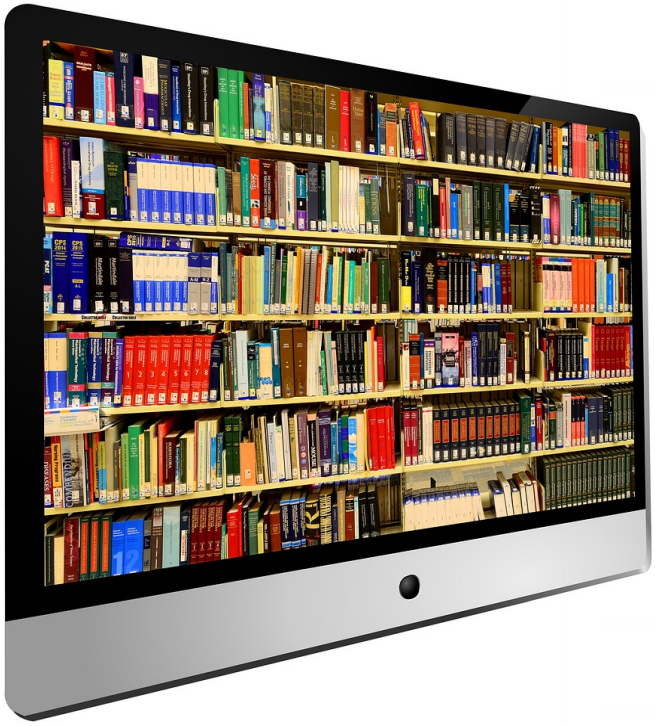
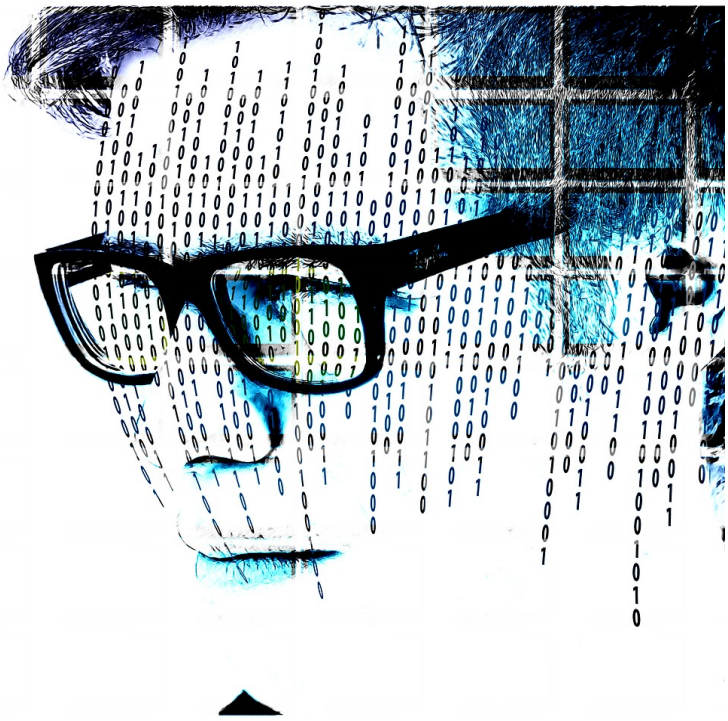
THE VIRTUAL MEET EXPERIENCE

2019 - 2020

HS VIRTUAL CHALLENGE MEET #1





COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2019-2020 Virtual Challenge Meet #1 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.7.25**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

Which of the following is equivalent to the expression `0b111100`?

A. 60_{10} B. 330_4 C. 74_8 D. $3C_{16}$ E. All are equivalent

QUESTION 2

What is output by the code to the right?

A. `35` B. `10`
C. `55`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
out.println(5+5 * 5+5);
```

QUESTION 3

What is output by the code to the right (e's indicate blank spaces)?

A. `5.35` B. `eee5.35`
C. `5.34` D. `eee5.34`
E. There is no output due to a compile error.

```
out.printf("%4.2f", 5.345);
```

QUESTION 4

What is output by the code to the right?

A. `That ain't it.Ight imma head out.`
B. `What it do What it does`
C. `R0shan.Ight Imma head out.`
D. `R0shan.`
E. There is no output due to a compile error.

```
String h="apluscompsci";
if(h.startsWith("aplus"))
  out.print("R0shan.");
else
  out.print("That ain't it.");
out.println("Ight Imma head out.");
```

QUESTION 5

What is output by the code to the right?

A. `True` B. `False` C. `false` D. `true`
E. There is no output due to a syntax error.

```
boolean a=true;
boolean b=false;
boolean c=a^b||b;
out.println(c&&a);
```

QUESTION 6

Which of the following fractions is equivalent to the decimal value outputted by the code to the right?

A. `π/3` B. `π/6` C. `π/2` D. `π/4`
E. There is no output due to an error.

```
double k=Math.PI/3;
double v=Math.cos(k);
double b=Math.asin(v);
out.printf("%.2f", b);
```

QUESTION 7

What is the output by the code to the right?

A. `3` B. `234`
C. `124` D. `14`
E. `4`

```
int y=8;
if(y%2==1)out.print(1);
if(y%3==1)out.print(2);
if(y%4==1)out.print(3);
if(y%7==1)out.print(4);
```

QUESTION 8

What is the output by the code to the right?

A. 15 B. 20
C. 18 D. 17
E. 12

```
int sum=2;
for(int h=0;h<13;h=(h+1)*3/2)
  sum+=3;
out.println(sum);
```

QUESTION 9

What is output by the code to the right?

A. 13 B. 12
C. 12.5 D. 12.0
E. There is no output due to an error.

```
int y=9;
y+=3.5;
out.println(y);
```

QUESTION 10

What is the output by the code to the right ?

A. 12 B. 13
C. 11
D. There is no output due to a runtime error
E. There is no output due to a syntax error.

```
int[] k= {1, 2, 3, 4, 5};
k[1]=k[2];
k[0]+=k[3]+=k[2]+k[4];
out.println(k[3]);
```

QUESTION 11

Which of the following values can be in `n.txt` so that the code runs without error?

A. 17
B. BeTheHyena
C. 23.43
D. All of the above.
E. None of the above.

```
File n=new File("n.txt");
Scanner sc=new Scanner(n);
String s=sc.next();
```

QUESTION 12

What is the output by the code to the right ?

A. 512 B. 256
C. 64 D. 128
E. 1

```
String s="*";
for(int y=0;y<8;y++)
  s+=s;
out.println(s.length());
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. III, IV, II, I B. III, I, II, IV
C. III, II, IV, I D. I, II, III, IV
E. IV, III, I, II

```
I. %
II. <=
III. []
IV. ?:
```

QUESTION 14

What is the output by the code to the right ?

A. 4 B. 16
C. 8 D. 64
E. 32

```
out.println(Float.SIZE);
```

QUESTION 15

What is the output by the code to the right ?

A. `[212, 7, 12, 17]`

B. `[7, 212, 12, 17]`

C. `[212, 7, 17]`

D. `[7, 12, 17]`

E. There is no output due to an error.

```
ArrayList<Integer>a;
a=new ArrayList<Integer>();
a.add(212);
a.add(12);
a.set(1, 7);
a.add(17);
out.println(a);
```

QUESTION 16

What is the output by the code to the right ?

A. `7` B. `1`

C. `6` D. `5`

E. There is no output due to a syntax error.

```
out.println(4|3|2&6|6|7^5);
```

QUESTION 17

What is the output by the line marked `//1` in the code to the right ?

A. `Brazy` B. `Crazy`

C. `Red` D. `BigHead`

E. There is no output due to an error.

QUESTION 18

What is output by the line marked `//2` in the code to the right?

A. `[Crazy, R0B0T, NutPay, Red]`
B. `[Crazy, NutPay, R0B0T, Red]`
C. `[R0B0T, Red]`
D. `[NutPay, Red]`
E. There is no output due to an error.

```
PriorityQueue<String>pq;
pq=new PriorityQueue<String>();
pq.add("BigHead");
pq.peek();
pq.add("NutPay");
pq.add("R0B0T");
pq.add("Red");
pq.add("Brazy");
pq.poll();
pq.add("Crazy");
out.println(pq.peek()); //1
pq.poll();
out.println(pq); //2
```

QUESTION 19

What is output by the code to the right?

A. `Hello World` B. `f c`
C. `HelloWorld`
D. Output cannot be determined until runtime.
E. No output due to an error.

```
String f="Hello";
char[] c= {'W','o','r','l','d'};
out.println(f+" "+c);
```

QUESTION 20

Which of the following is a valid Java identifier?

A. `1776` B. `#123` C. `Who?What?` D. `Ca$hMoney` E. None of the above.

QUESTION 21

What is the output by the code to the right ?

A. `8 17 7 30`
B. `30 7 17 8`
C. `17 7`
D. `8 30`
E. There is no output due to a runtime error.

```
ArrayList<Integer> x;
x = new ArrayList<Integer>();
x.add(8);
x.add(17);
x.add(7);
x.add(30);
x.forEach(n -> { if (n%2 == 0)
System.out.print(n + " "); });
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right?

A. `super();`

B. `super(n);`

C. `super(yy);`

D. `super(yy, n);`

E. `super(n, yy);`

QUESTION 23

Assuming that **<1*>** is filled correctly, what is the output by `//1` ?

A. `5`

B. `YES`

C. `4`

D. `8`

E. `15`

QUESTION 24

Assuming that **<1*>** is filled correctly, what is the output by `//2` ?

A. `12`

B. `13`

C. `15`

D. `14`

E. `NO`

QUESTION 25

Assuming that **<1*>** is filled correctly, what is the output by `//3` ?

A. `21`

B. `24`

C. `15`

D. `18`

E. `MAYBE`

```
class A{
  private int y;
  String name;
  public A(String n) {
     y=5;
     name=n;
  }
  public int getY() {
     return y;
  }
}
class B extends A{
  private int y;
  String name;
  public B(int yy,String n) {
     <1*>;
     y=yy;
  }
  public int getY() {
     return super.getY()+y*2;
  }
}
///////////client code//////////////
A a=new A("YES");
A aa=new B(4,"NO");
B b=new B(8, "MAYBE");
out.println(a.getY()); //1
out.println(aa.getY()); //2
out.println(b.getY()); //3
```

QUESTION 26

What is output by the code to the right?

A. `false true`

B. `false false`

C. `true false`

D. `true true`

E. There is no output due to a runtime error.

```
String s="Somehow1Manage";
out.print(s.matches("\\w+")+" ");
out.print
 (s.matches("[A-Z][a-z]+[1-3]\\D+"));
```

QUESTION 27

What is output by the code to the right?

A. 5 B. 3

C. 4 D. 6

E. There is no output due to a syntax error.

```
int h=212;
int i=Integer.bitCount(h);
out.println(i);
```

QUESTION 28

What can replace **<1*>** and **<2*>** in the code to the right so that **<1*>** adds to the stack and **<2*>** removes?

	<1*>	**<2*>**
A.	push	pop
B.	add	peek
C.	add	remove
D.	push	peek
E.	push	remove

QUESTION 29

What is output by the code to the right?

A. [212, Hello, It's Me, I was Wondering]
B. [212, Hello, I was Wondering, If after all these]
C. [212, I was Wondering, years youd like to meet]
D. [212, Tha Thing, Hello, It's Me]
E. There is no output due to an error.

```
Stack<String> s;
s=new Stack<String>();
s. <1*> ("212");
s. <1*> ("Tha Thing");
s. <2*> ();
s. <1*> ("Hello");
s. <1*> ("It's Me");
s. <1*> ("I was Wondering");
s. <1*> ("If after all these");
s. <1*> ("years youd like to meet");
s. <2*> ();
s. <2*> ();
out.println(s);
```

QUESTION 30

What of the following is demonstrated by method `fun`?

A. Pythagorean Theorem

B. Dynamic Programming

C. Dijkstra's algorithm

D. A logarithmic sequence

E. The Fibonacci Sequence

```
double fun(int a, int b) {
  int c=a*a+b*b;
  return Math.sqrt(c);
}
```

QUESTION 31

What is the value of the postfix expression to the right(use integer division)?

A. 4 B. 22 C. 26 D. 20 E. 8

```
7 8 + 2 * 4 + 3 / 2 1 1 4 3 - * * * *
```

QUESTION 32

Which of the following lines in the code to the right will cause a syntax error?

A. //1 B. //2

C. //3 D. More than one of the above.

E. None of the above.

QUESTION 33

Assuming any error has been corrected, what is output by the code to the right ?

A. 0 B. 1

C. 2 D. 3

E. There is no output due to a runtime error.

```
int y, x=5;  //1
switch(x) {
case 5: y=2; //2
case 4: y=6;
default: y=3;
}
out.print(x/y); //3
```

QUESTION 34

What could replace **<1*>** in the code to the right, so the code will compile without error?

A. `String`
B. `E`
C. `Object`
D. More than one of the above.
E. None of the above.

QUESTION 35

What is output by the following line of code?
`out.println(Structure.val("S0M3B0DY2L0V3"));`

A. `96`
B. `97`
C. `105`
D. `106`
E. There is no output due to an error.

QUESTION 36

What could replace **<2*>** in the client code to initialize a `Structure` object called `s` of size 15?

A. `Structure<String> s=new Structure<String>(15);`
B. `Structure s=new Structure(15);`
C. `Structure<int[]> s=new Structure<int[]>(15);`
D. None of the above.
E. All of the above.

QUESTION 37

What is output by the client code to the right ?

A. `gotem GOODBYE POST 212 !@#$`
B. `gotem !@#$ GOODBYE POST 212`
C. `!@#$ 212 POST gotem GOODBYE`
D. `!@#$ POST gotem GOODBYE 212`
E. There is no output due to an error.

QUESTION 38

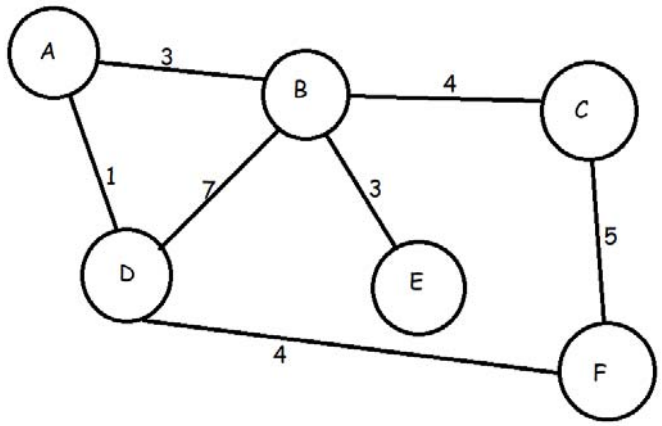
Which of the following data structures is implemented by the code to the right ?

A. Linked List
B. Priority Queue
C. Hashtable
D. Stack
E. Set

```
class Structure<<1*>>{
  LinkedList<String>[]struct;
  int size;
  public static int val(String h) {
          int n=0;
          for(int y=0;y<h.length();y++)
if(Character.isAlphabetic(h.charAt(y)))
                n+=h.charAt(y)-'A'+1;
          return n;
  }
  public Structure(int h) {
          size=h;
          struct=new LinkedList[size];
          for(int k=0;k<size;k++)
  struct[k]=new LinkedList<String>();
  }
  public boolean add(String h) {
          int g=val(h)%size;
          if(struct[g].contains(h))
            return false;
          struct[g].add(h);
          return true;
  }
  public boolean remove(String h) {
          int g=val(h)%size;
          if(!struct[g].contains(h))
            return false;
          struct[g].remove(h);
          return true;
  }
  public String toString() {
          String fin="";
          for(int y=0;y<size;y++)
  for(int x=0;x<struct[y].size();x++)
            fin+=struct[y].get(x)+" ";
          return fin.trim();
  }
}
//////////client code/////////////////
<2*>
s.add("!@#$");
s.add("HELLO");
s.add("GOODBYE");
s.add("POST");
s.remove("HELLO");
s.add("gotem");
s.add("212");
out.println(s);
```

QUESTION 39

What is the minimum cost path from node A to F in the following graph?



QUESTION 40

What is the most simplified form of the following boolean algebra expression ?

```
A && B && C || A && B || A && C
```

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets

Computer Science - Student Answer Sheet

1) _____
2) _____
3) _____
4) _____
5) _____
6) _____
7) _____
8) _____
9) _____
10) _____

11) _____
12) _____
13) _____
14) _____
15) _____
16) _____
17) _____
18) _____
19) _____
20) _____

21) _____
22) _____
23) _____
24) _____
25) _____
26) _____
27) _____
28) _____
29) _____
30) _____

31) _____
32) _____
33) _____
34) _____
35) _____
36) _____
37) _____
38) _____
39) _____
40) _____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2019-2020 Virtual Challenge Meet #1
Computer Science - KEY

1) E
2) A
3) A
4) C
5) D
6) B
7) E
8) D
9) B
10) A

11) D
12) B
13) B
14) E
15) C
16) A
17) A
18) B
19) D
20) D

21) D
22) B
23) A
24) B
25) A
26) D
27) C
28) A
29) A
30) A

31) B
32) E
33) B
34) D
35) B
36) E
37) C
38) C
39) 5
40) A && (B || C)

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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